Current Affairs 02-09-2026

Credit - Drishti Ias

C-DOT Unveils 14 Indigenous Quantum Products for Secure Communication Networks

Source: TH

Why in News?

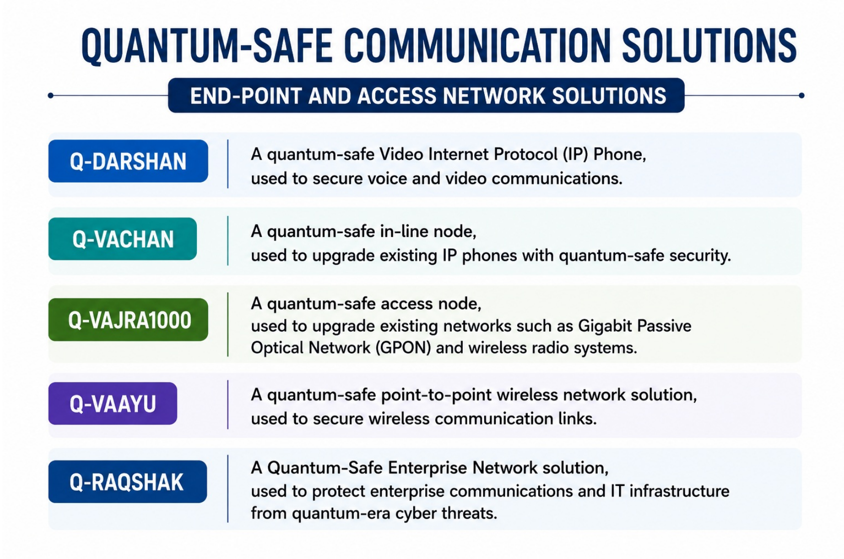
The **Centre for Development of Telematics (C-DOT)** , under the **Department of Telecommunications (DoT), Ministry of Communications** , unveiled **14 indigenous quantum products** to strengthen the security of India’s communication networks, focusing on **Quantum Key Distribution (QKD)** and **Post-Quantum Cryptography (PQC)** during its **43 rd Foundation Day celebrations** .

What are the Key Quantum Technologies Developed by C-DOT?

- **Quantum Key Distribution (QKD) Systems:**
  - **Q-AKSHAY CD:** A compact fibre-based **Quantum Key Distribution (QKD)** system using **Coherent One-Way (COW)** and **Differential Phase Shift (DPS)** protocols, **used for secure quantum key generation and distribution over fibre networks** .
  - **Q-AKSHAY MD:** A fibre-based **QKD** system using the **Measurement Device Independent (MDI)** protocol, **used for secure key distribution with enhanced protection against measurement-device vulnerabilities** .
- **Quantum-Safe Encryptors:**
  - **Q-VIKRAM & Q-PARAKRAM:** Defence-grade quantum-safe encryptors using **Post-Quantum Cryptography (PQC)** algorithms, operating at **Layer 2/3 (Data Link and Network Layers) of the OSI (Open Systems Interconnection) model** , used to secure strategic and defence communications.
  - **Q-AMOGH:** A high-capacity optical encryptor providing up to **200 Gbps** quantum-safe encryption at Layer 1, **used to secure high-speed optical communication networks** .
  - **Q-MAHASETU:** A commercial-grade quantum-safe encryptor supporting up to **40 Gbps** at Layer 2/3, **used to protect enterprise and critical communication networks** .
  - **Q-SETU:** A quantum-safe encryptor providing up to **80 Mbps** at Layer 3, **used to secure commercial and enterprise data communications** .
    - It incorporates **National Institute of Standards and Technology (NIST) PQC algorithms** to protect data against emerging quantum-era threats.
- **Critical Quantum Hardware Components:**

- **C-SPD:** Single-Photon Detector, used to detect single-photon signals in quantum communication systems.
- **C-RD:** Wideband Radio Frequency (RF) Driver, used to drive intensity and phase modulators in quantum communication systems.

- **End-Point and Access Network Solutions:**



OSI (Open Systems Interconnection) Model

- It is a **7-layer framework** that explains how data moves between devices over a network:
  - **Layer 1 – Physical:** Transmits raw data as physical signals.
  - **Layer 2 – Data Link:** Handles communication between directly connected devices.
  - **Layer 3 – Network:** Handles **routing and addressing** of data between networks.
  - **Layer 4 – Transport:** Ensures reliable **end-to-end data delivery** .
  - **Layer 5 – Session:** Establishes and manages communication sessions.
  - **Layer 6 – Presentation:** Handles **data formatting, encryption and compression** .
  - **Layer 7 – Application:** Provides network services directly to applications.

What is Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC)?

- **Quantum Computing:** It is an emerging field that uses the **principles of quantum mechanics** to solve certain complex problems beyond the capabilities of classical computers.
  - By harnessing quantum phenomena, quantum computers could perform specific calculations in minutes or hours that would take conventional supercomputers thousands of years.
- **Quantum Key Distribution (QKD) Network:** QKD is a highly secure way to share secret encryption keys between two users using quantum particles (photons). It ensures that communication remains safe from hacking.
  - If anyone tries to intercept the key, the quantum particles get disturbed (due to the observer effect and no-cloning rule), which immediately alerts the users about the intrusion.
- **Post-quantum cryptography (PQC):** It is a type of encryption designed to stay secure against attacks

from powerful quantum computers.

QKD	PQC
Based on **quantum mechanics**	Based on **classical mathematics/cryptography**
Securely **generates and distributes encryption keys**	Provides **quantum-resistant encryption algorithms**
Generally requires specialised **quantum communication infrastructure**	Can be implemented on **existing communication networks**
Example: **Q-AKSHAY CD, Q-AKSHAY MD**	Example: **Q-SETU, Q-MAHASETU, Q-VIKRAM**

Centre for Development of Telematics (C-DOT)

- **About:** C-DOT established on **24 th August 1984** by the Government of India as a telecom technology development centre under the **Department of Telecommunications (DoT)** .
    - C-DOT is a premier **indigenous telecom R&D institution** , instrumental in India's telecom revolution, particularly in **rural telephony and network digitisation** .
- **Major Technologies:** Developed solutions in **Next Generation Networks (NGN), Terabit Routers, XGS-PON, DWDM, Wi-Fi, Network Management Systems (NMS), cybersecurity and telecom software** .
- **IoT/M2M:** Developed the **oneM2M standards-based Common Services Platform (CCSP)** to support **Internet of Things (IoT), Machine-to-Machine (M2M), Smart Cities and Industrial IoT** .
- **Emerging Technologies:** C-DOT is also working on **4G/5G, Artificial Intelligence (AI), quantum communication and 6G** technologies.
- **Technology Transfer:** Its **Technology Transfer model** has helped create a domestic ecosystem of **telecom equipment manufacturers and component suppliers** .
- **Significance:** C-DOT advances **Atmanirbhar Bharat, Digital India and indigenous telecom manufacturing** by developing technologies suited to India's needs.
    - Its technologies have contributed to **BharatNet** , the national optical fibre network aimed at expanding broadband connectivity to rural areas.

Frequently Asked Questions (FAQs)

1. **What is Quantum Key Distribution (QKD)?**
   QKD uses **quantum-mechanical principles** to securely generate and distribute encryption keys, with interception detectable through changes in quantum states.
2. **What is Post-Quantum Cryptography (PQC)?**
   PQC uses **classical cryptographic algorithms** designed to remain secure against attacks by future quantum computers.
3. **What is the difference between QKD and PQC?**
   QKD uses quantum communication infrastructure for secure key distribution , while **PQC uses quantum-resistant mathematical algorithms** that can generally work over existing networks.
4. **What are Q-AKSHAY CD and Q-AKSHAY MD?**
   Both are **fibre-based QKD systems** developed by C-DOT. Q-AKSHAY CD uses **COW and DPS protocols** , while Q-AKSHAY MD uses the **Measurement Device Independent (MDI) protocol** .
5. **Why are C-DOT’s indigenous quantum products significant for India?**
   They strengthen **Atmanirbhar Bharat, cybersecurity and strategic autonomy** by providing indigenous quantum-safe solutions for **defence, telecom, enterprise and critical communication networks** .

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/WOxz5mSHOKk**]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. 'X', born in the UK, was conferred the Nobel Prize in 2025. He was a professor in an American university when this prize was announced. Identify 'X':(2026)

(a) Michel H. Devoret

(b) Richard Robson

(c) John Clarke

(d) Joel Mokyr

Ans: (C)

Q. Which one of the following is the context in which the term "qubit" is mentioned? (2022)

(a) Cloud Services
(b) Quantum Computing
(c) Visible Light Communication Technologies
(d) Wireless Communication Technologies

Ans: (b)

Mains

Q. “The emergence of the Fourth Industrial Revolution (Digital Revolution) has initiated e-Governance as an integral part of government”. Discuss. **(2020)**

India's 1st "Green Forge" Complex Inaugurated

Source: PIB

India’s first-of-its-kind **Green Forge Complex** was inaugurated at **National Agri-food & Biomanufacturing Institute (BRIC-NABI)** (an Autonomous Institute of the Department of Biotechnology) in **Mohali** , marking a major step in **agricultural biotechnology** and **genetically modified (GM) crop research** .

- **Context of the Launch:** The facility was inaugurated during the **National Conclave on BioE3 (Biotechnology for Economy, Environment and Employment), 2026** reflecting India's shift toward high-performance biomanufacturing.
- **'Green Forge' Facility:** It provides **advanced simulation chambers** that can precisely regulate light, humidity, and temperature.
    - This allows researchers to cultivate and experiment with **GM crops** to assess their suitability across varied and controlled environmental conditions before field deployment.
- **Technological Integration:** The facility highlights the convergence of biotechnology with **advanced simulation and Artificial Intelligence (AI)** , accelerating complex research in both agricultural and medical sciences.
- **Surge in India's Bioeconomy:** India’s **bioeconomy** has expanded from USD 10 billion in 2014 to approximately USD 200 billion, driven by private sector engagement and an enabling policy environment.
- **Infrastructure Push:** Under **BioE3 (Biotechnology for Economy, Environment and Employment) 2024 policy** , the focus is shifting towards **industry-ready manufacturing** and translating research into products and technologies.
    - The government is developing **22 biomanufacturing hubs** and a **National Bio-Foundry Network comprising eight specialized facilities** to support start-ups, SMEs, and the translation of research into commercial products.
    - The **Department of Biotechnology (DBT)** is actively supporting around **38 Carbon Capture and Utilisation (CCU) projects** to capture emissions from hard-to-abate sectors like cement and steel through integrated biomanufacturing.
- **Emerging Biotech Frontiers:** The initiative is expanding into **sustainable aviation fuels (SAF)** , the I3 PhD programme, and **Space Biotechnology** —including collaborations with **ISRO on myogenesis and cyanobacteria** growth experiments in space.
- **Waste-to-Wealth:** BRIC-NABI is heavily integrating agri-biotechnology with bioprocess engineering to convert **agricultural waste into high-value nutraceuticals** (products derived from food sources that provide extra health and medical benefits besides basic nutrition) **and biomaterials .**
- **Strategic Alignment:** The overall transformation toward a low-carbon, circular bioeconomy is directly aligned with India's long-term climate and economic goals of achieving **Net Zero by 2070** and **Atmanirbhar Bharat** .

Read more: Unlocking India's Bioeconomy Potential

Rani Avanti Bai Lodhi

Source: TOI

Defence Minister **Rajnath Singh** paid tribute to **Rani Avanti Bai Lodhi** , describing her patriotism and bravery as values that transcend social, caste and gender boundaries.

Rani Avanti Bai Lodhi

- **Birth and Family:** Rani Avanti Bai Lodhi was born on **16th August 1831** in **Manakhedi village of Seoni district, Madhya Pradesh** . Her father, **Rao Jujhar Singh** , was a zamindar, and her mother was **Krishna Bai** .
- **Training and Abilities:** From childhood, she developed skills in **horse riding, sword fighting and administration,** which later helped her in leading resistance against British rule.
- **Marriage and Role in Ramgarh State:** She married **Vikramaditya Singh** , son of **King Lakshman Singh of Ramgarh princely state** . After Vikramaditya Singh became king but remained ill due to health issues, Rani Avanti Bai took charge of the administration. The couple had two sons — **Aman Singh and Sher Singh** .
- **Resistance against British:** British authorities brought Ramgarh under the **Court of Wards system,** declaring the king incapable of ruling and his heirs minors. Rani Avanti Bai opposed British control, **removed British-appointed officials from Ramgarh** and took over administration.
- **Mobilising Farmers and Local Support:** She refused to accept British administrative control and encouraged farmers not to follow British revenue-related orders. This increased her popularity among the people.
- **Call for Armed Resistance:** During the **Revolt of 1857** , she organised resistance against British rule and sent letters along with **two black bangles** to local rulers, zamindars, and landlords, urging them either to fight the British or accept defeat by wearing bangles.
- **Battle of Kheri (1857):** On **23rd November 1857** , Rani Avanti Bai and her forces defeated the British army led by **Mandla Deputy Commissioner Waddington** in the **Battle of Kheri** , demonstrating her ability to organise armed resistance against British forces.
- **Final Resistance and Martyrdom:** In **March 1858** , British forces launched a major attack against Rani Avanti Bai and her troops. When surrounded by the enemy and facing certain capture, she chose self-sacrifice rather than surrender, becoming a symbol of resistance against colonial rule.
- **Symbol of Courage and Patriotism:** Rani Avanti Bai Lodhi is remembered for her **courage, leadership, self-respect and commitment to the freedom struggle** . Her role highlights the contribution of women and regional leaders in India’s anti-colonial movement.
- **Recognition:** Her legacy continues to be honoured through memorials, institutions and public celebrations, including initiatives such as the **Rani Avanti Bai Lodhi Women’s Battalion of the Uttar Pradesh Provincial Armed Constabulary (PAC) at Badaun** .

Read More: Revolt of 1857

Cyber-Physical Systems

Source: PIB

Why in News?

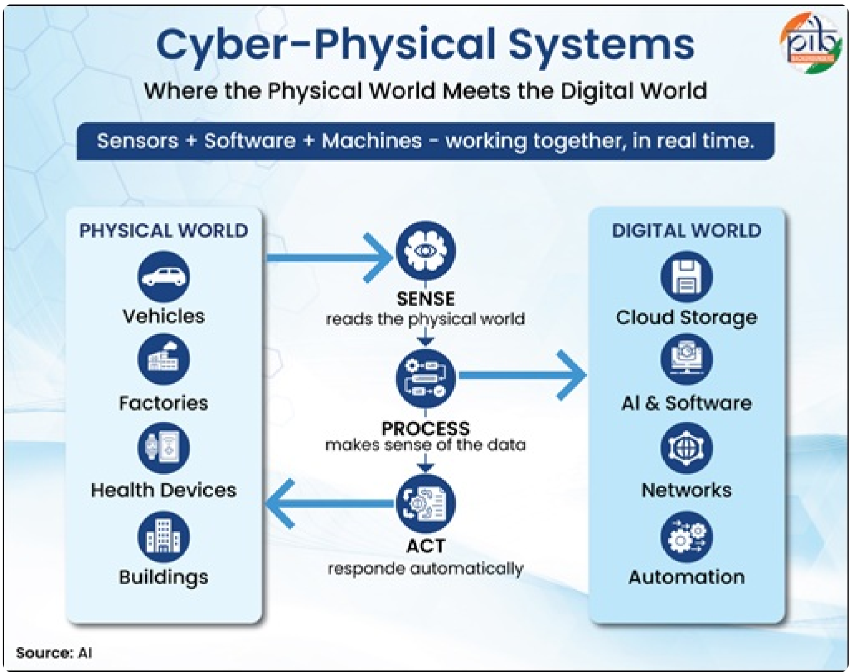
The Government has highlighted the growing importance of **Cyber-Physical Systems (CPS)** in building a technology-driven future. India is developing a strong CPS ecosystem through the **National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS) .**

What are the Key Facts About Cyber-Physical Systems (CPS)?

- **About:** Cyber-Physical Systems (CPS) are intelligent systems that integrate the physical and digital worlds through sensors, computing systems, software, communication networks and actuators. They enable machines to sense, analyse, decide and respond to real-time conditions.
- **Working of CPS:** CPS connects **computation, communication, and physical processes** through a continuous feedback loop. Sensors collect data from the environment, computing systems analyse the information, and actuators perform necessary actions.
- **Examples of CPS Applications:**

- **Autonomous Vehicles:** Driverless cars use sensors and AI-based systems to navigate roads with minimal human intervention.
- **Smart Manufacturing:** Smart factories use CPS for automated monitoring, optimisation and Industry 4.0 applications.
- **Healthcare:** Intelligent medical devices continuously monitor patients and support timely medical interventions.
- **Smart Buildings:** Automated systems regulate lighting, temperature, and energy use based on real-time conditions.

- **Significance:** CPS enables the development of **intelligent, connected, and responsive systems** , supporting advancements in areas such as automation, healthcare, transportation, manufacturing, and smart infrastructure.

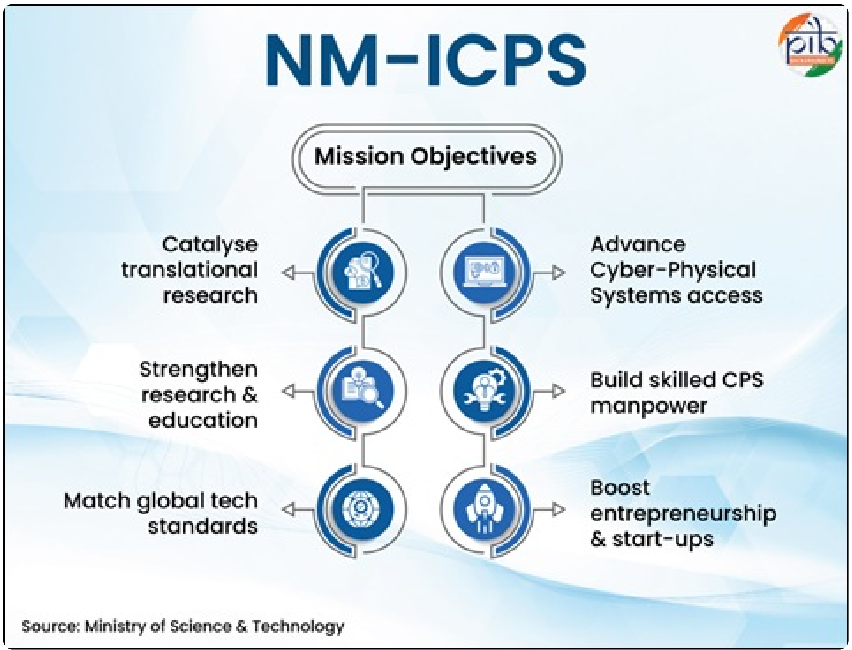


What are the Key Facts About National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)?

- **About:** The **Department of Science and Technology (DST)** is implementing NM-ICPS to develop indigenous CPS capabilities in India. The mission was approved by the **Union Cabinet in 2018** with an **outlay of ₹3,660 crore for nine years** .
- **Objective:** The mission provides a national framework for advancing **CPS research, innovation, and applications** across priority sectors by integrating **academia, industry, government agencies, and international organisations** .
- **Innovation Ecosystem:** NM-ICPS supports the complete innovation cycle, including **research and development (R&D), translational research, product development, and start-up incubation** , enabling the conversion of scientific knowledge into practical solutions.
- **Centres of Excellence:** The mission is establishing **world-class interdisciplinary Centres of Excellence** across the country to promote advanced CPS research, technology development, specialised expertise, and policy support.
- **Technology Development:** NM-ICPS focuses on developing **national-priority technologies, core competencies, and skilled human resources** in emerging CPS domains.

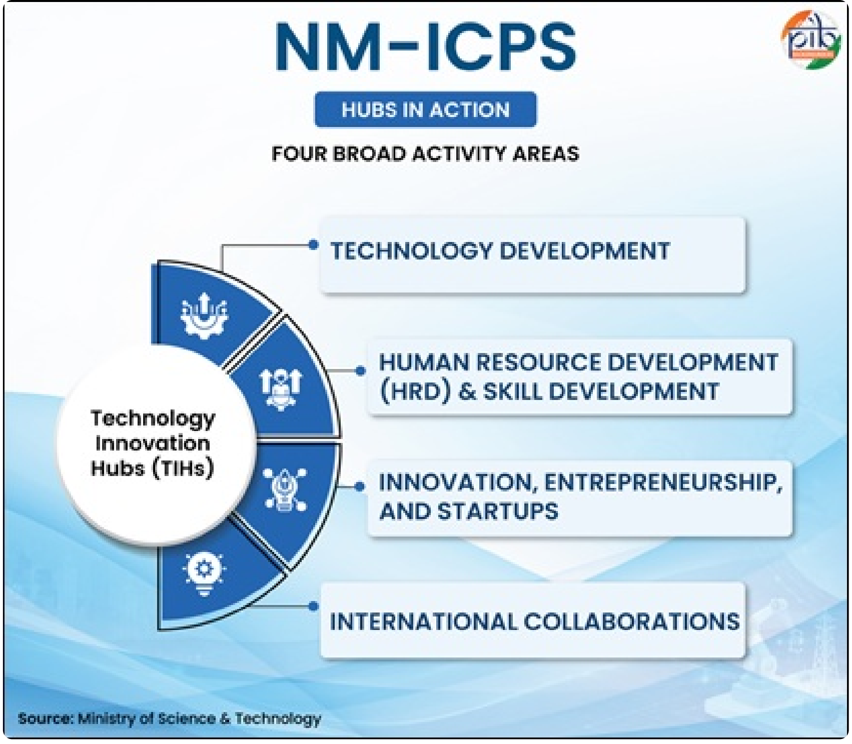
Achievements of NM-ICPS (as of August 2026)

- **Technology Development:** The mission has enabled **1,146 technologies** and supported the development of **1,329 technology products** across sectors such as **agriculture, healthcare, mining, cybersecurity, advanced communication, and positioning systems** .
- **Skill Development:** Through the **CHANAKYA Fellowship Programme** , NM-ICPS supports undergraduate, postgraduate, PhD, and postdoctoral researchers working on CPS challenges.
- **Start-up Ecosystem:** The mission has supported **1,136 start-ups and spin-offs** , generating more than **24,000 jobs** by promoting research-based entrepreneurship.
- **Global Collaboration:** NM-ICPS has established **200 international collaborations** to strengthen India's participation in the global CPS technology ecosystem.



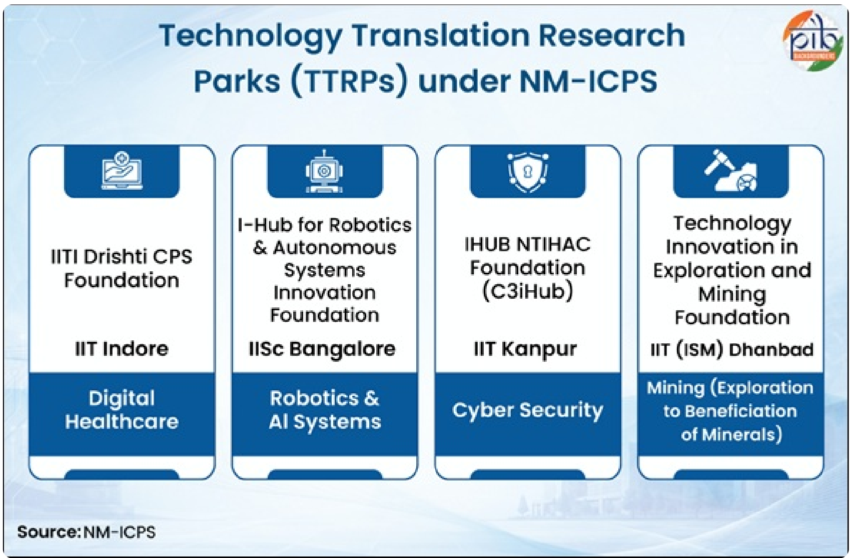
Technology Innovation Hubs (TIHs)

- **About:** Under **NM-ICPS** , the Government has established **25 Technology Innovation Hubs (TIHs)** as **Section 8 companies** in leading academic institutions across India. These hubs focus on **technology development, human resource development, innovation and start-up support, and international collaborations** .
- **Role of TIHs:** TIHs bridge the gap between research and real-world applications by developing technology platforms, converting research into products, nurturing technology-based start-ups, and working with government agencies to solve national challenges.



Technology Translation Research Parks (TTRPs)

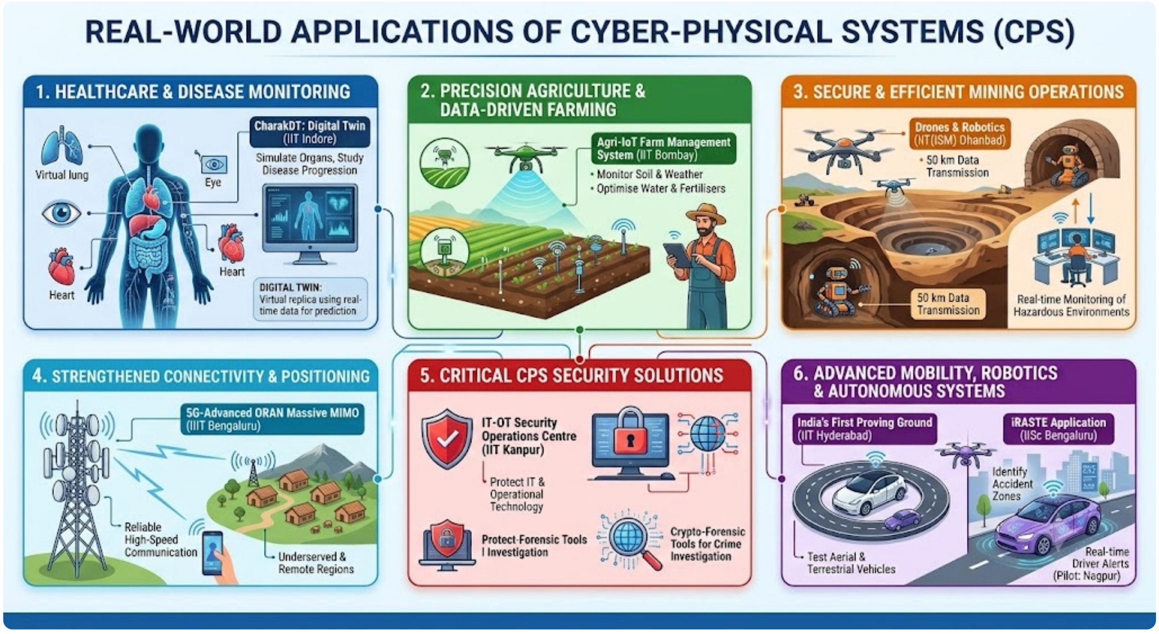
- **About:** To accelerate commercialisation of CPS technologies, **four high-performing TIHs were selected in 2025 as TTRPs** at:
  - **IIT Kanpur:** Cybersecurity
  - **IISc Bengaluru:** Robotics and AI Systems
  - **IIT (ISM) Dhanbad:** Mining Technologies
  - **IIT Indore:** Digital Healthcare
- **Objective:** TTRPs aim to accelerate the transition of CPS innovations from laboratories to **deployable technologies and commercial products** .



BharatGen

- **About:** Supported under **NM-ICPS** , **BharatGen** is developing **sovereign AI capabilities** tailored to Indian languages and contexts. It is led by **IIT Bombay** along with a consortium of academic institutions.
- **Multilingual AI Capability:** BharatGen develops **text, speech, and vision-language technologies** to enable human-machine interaction across India’s linguistic diversity.

- **AI Models Developed:**
  - **Param-2:** A foundation model with **17 billion parameters** supporting all **22 Scheduled Indian languages** .
  - **Shrutam:** Speech-to-text model supporting **12 Indian languages** .
  - **Sooktam:** Text-to-speech model supporting **12 Indian languages** .
  - **Patram:** Enables multilingual access to complex Indian documents through the **DocBodh framework** .
- **Sector-Specific AI Models:** BharatGen has developed domain-focused models such as:
  - **Ayur Param** for Ayurveda,
  - **Agri Param** for agriculture,
  - **Legal Param** for the Indian legal sector.



Conclusion

Cyber-Physical Systems represent the next phase of technological evolution by integrating digital intelligence with physical infrastructure. Through **NM-ICPS** , India is building future-ready capabilities in AI, robotics, cybersecurity and advanced technologies, paving the way for smarter systems, innovation-led growth and a **technology-driven, self-reliant Viksit Bharat** .

Drishti Mains Practice Question

Q. Discuss the role of India’s National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS) in building indigenous capabilities in Cyber-Physical Systems (CPS).

Frequently Asked Questions (FAQs)

1. What are Cyber-Physical Systems (CPS)?

Cyber-Physical Systems are intelligent systems that integrate the **physical and digital worlds** using sensors, software, computing systems and communication networks. They enable machines to **sense, analyse and respond** to real-time conditions.

2. What are the main components of CPS?

Sensors, Computing systems, Communication networks and Actuators.

3. What are some examples of Cyber-Physical Systems?

- Autonomous vehicles.
- Smart factories.
- Digital healthcare devices.
- Smart buildings.
- Fitness trackers and cleaning robots.

4. What is the National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)?

NM-ICPS is a Government of India initiative implemented by the **Department of Science and Technology (DST)** to develop indigenous CPS capabilities.

5. What is the objective of NM-ICPS?

To promote research, innovation and application of CPS technologies. To connect academia, industry, government and international organisations for technology development.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/HN9bjJfzlIl**]

UPSC Civil Services Examination, Previous Year Questions

Q1. In India, under cyber insurance for individuals, which of the following benefits are generally covered, in addition to payment for the loss of funds and other benefits? (2020)

1. Cost of restoration of the computer system in case of malware disrupting access to one's computer
2. Cost of a new computer if some miscreant wilfully damages it, if proved so
3. Cost of hiring a specialised consultant to minimise the loss in case of cyber extortion
4. Cost of defence in the Court of Law if any third party files a suit

Select the correct answer using the code given below:

(a) 1, 2 and 4 only

(b) 1, 3 and 4 only

(c) 2 and 3 only

(d) 1, 2, 3 and 4

Ans: (b)

Q2. In India, it is legally mandatory for which of the following to report on cyber security incidents? (2017)

1. Service providers
2. Data centres
3. Body corporate

Select the correct answer using the code given below:

(a) 1 only

(b) 1 and 2 only

(c) 3 only

(d) 1, 2 and 3

Ans: (d)

Mains

Q. What are the different elements of cyber security ? Keeping in view the challenges in cyber security, examine the extent to which India has successfully developed a comprehensive National Cyber Security Strategy. **(2022)**

Vultures in India

Source: TH

A study published in the **Journal of Threatened Taxa** has highlighted that relocating a **livestock carcass dump away from high-tension power lines** near **Sudhauwala in Dehradun, Uttarakhand** helped reduce the risk of vulture electrocution.

- **About Vultures:** Vultures are **large scavenger birds that feed on dead animals** and act as **nature's sanitation workers.** They help maintain ecosystem cleanliness by removing carcasses, supporting nutrient cycling and reducing the spread of diseases.
  - Their **highly acidic digestive system helps destroy harmful pathogens** present in carcasses, reducing the risk of diseases such as anthrax and botulism.
- **Vulture Diversity in India:** Around 23 species of vultures are found globally. India is home to 9 species of vultures, including the **White-rumped Vulture, Indian Vulture, Slender-billed Vulture, Himalayan Griffon , Eurasian Griffon, Egyptian Vulture, Red-headed Vulture , Cinereous Vulture and Bearded Vulture** .
- **Indicator Species:** Vultures are considered **indicators of ecosystem health** because their decline reflects wider environmental issues such as pollution, food scarcity and habitat degradation.
- **Population Decline:** Vulture populations in South Asia, particularly **India, Pakistan and Nepal** , declined sharply due to human-induced threats. The widespread use of veterinary **diclofenac (a non-steroidal anti-inflammatory drug, NSAID)** was the major cause, resulting in a decline of over **97% in some regions** . India banned veterinary diclofenac in **2006** .
- **Emerging Threats:** Vultures face additional threats from **other NSAIDs, food scarcity, habitat loss, poisoning and electrocution from high-tension power infrastructure** .
- **Conservation Measures:** India has undertaken measures such as the **Vulture Conservation Breeding Programme,** promotion of safer alternatives like meloxicam, establishment of Vulture Safe Zones, and habitat-sensitive management practices to support vulture recovery.
- **Vulture Awareness Day:** The **first Saturday of September** is recognised annually as **Vulture Awareness Day** to highlight the ecological importance of vultures and promote their conservation.

Sr. No.	Name of the Vulture Species	IUCN status	Pictorial Representation
1.	Oriental White-backed Vulture (Gyps Bengalensis)	**Critically Endangered**	
2.	Slender-billed Vulture (Gyps Tenuirostris)	**Critically Endangered**	
3.	Long-billed Vulture (Gyps Indicus)	**Critically Endangered**	
4.	Egyptian Vulture (Neophron Percnopterus)	**Endangered**	
5.	Red-Headed Vulture (Sarcogyps Calvus)	**Critically Endangered**	
6.	Indian Griffon Vulture (Gyps Fulvus)	**Least Concerned**	
7.	Himalayan Griffon (Gyps Himalayensis)	**Near Threatened**	
8.	Cinereous Vulture (Aegypius Monachus)	**Near Threatened**	
9.	Bearded Vulture or Lammergeier (Gypaetus Barbatus)	**Near Threatened**	

Read More: Vultures at Risk in Protected Areas